

Documenting Changes in 02_clean_validate/README.Rmd

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0.1 Purpose

Provide a human-readable summary of scripts in `R/02_clean_validate`, including their purpose, input/output expectations, status, and key changes over time. This supports reproducibility and onboarding of new contributors.

0.2 Overview

This section documents the current structure and status of scripts responsible for processing files in `data/raw`. Each script is version-controlled and tied to a specific covariate or source dataset.

Table 1: Script Index

File Name	Description	Output Folder	Notes / Status
02a_merge_nhdplus_hr_flowlines.R	Merge NHDPlus HR flowlines	data/processed/nhdphr/	CRS standardized (EPSG:5070)
02b_merge_nhdplus_hr_catchments.R	Merge NHDPlus HR catchments	data/processed/nhdphr/	CRS standardized (EPSG:5070)
02c_validate_spatial_metadata.R	Validate CRS and resolution of spatial files.	data/log/	
02d_extract_metadata_from_script_names.R	Generate a CSV tracker in covariate_source_inventory.Rmd	data/log/	Output is a table

0.3 Summary of Key Changes

0.3.1 Standardized Folder Paths

- Output paths outside of functions use `here::here("data", "processed")`

0.3.2 Consistent CRS

- Spatial outputs reprojected to EPSG:5070 (CONUS Albers Equal Area)

0.3.3 Modular Design

- Each script targets a single source or covariate
- Utility functions moved to `R/utils/`

0.3.4 Logging & QA

- Retry attempts, download failures, and chunking behavior logged to `.csv` where applicable
- Scripts include `{cli}` alerts for status tracking

0.3.5 Metadata Documentation

- Consistent headers at the top of each script.
- Script headers ready to be cross-linked to relevant milestone reports